

2.3.1. Measurement of Direct Radiation using Pyrheliometer

A pyrheliometer is an instrument which measures the direct beam solar radiation. It is an instrument which collimates the radiation to determine the beam intensity as a function of incident angle. To measure the direct solar radiation correctly, its receiving surface must be arranged to be normal to the solar direction i.e., a line joining the sun and receiver. For this reason, the instrument is usually mounted on a sun-tracking device called an *equatorial mount*.

In this device, a sensor disc is located at the base of a tube whose axis is aligned with the direction of the sun rays. Thus, the diffuse radiation is essentially blocked from the sensor surface. Most of the pyrheliometers operate on the principle of thermopile effect. The diffuse component is avoided by installing a collimator tube over the sensor with a circular cone angle of about 5° .

Three types of pyrheliometers are used to measure the normal incident beam radiation. They are as follows:

1. Angstrom pyrheliometer
2. Abbot silver disc pyrheliometer
3. Eppley pyrheliometer.

1. Angstrom compensation pyrheliometer

In this type, a thin blackened shaded manganin strip (size $20 \times 2 \times 0.1 \text{ mm}$) is heated electrically until it is at the same temperature as a similar strip which is exposed to solar radiation.

radiation. Figure 2.7 shows the electric circuit of this pyrliometer. Under steady state conditions (both strips at identical temperature), the energy used for heating is equal to the absorbed solar energy. The thermocouples on the back of each strip is connected in opposition through a sensitive galvanometer to test the equality of temperature.

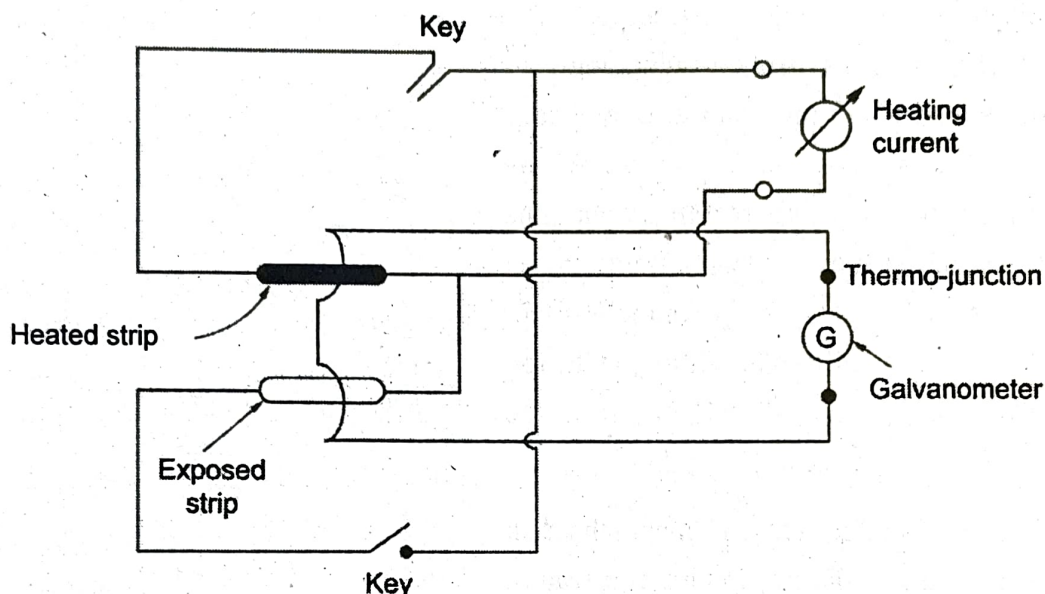


Figure 2.7 Electric circuit of Angstrom compensation pyrliometer

The energy H of direct radiation is calculated by using the formula.

H_{DN} = Direct radiation incident on the area normal to sun rays

$$H_{DN} = K I^2$$

where

I = Heating current in amperes

K = Dimension and instrument constant = $\frac{R}{W \alpha}$

R – The resistance per unit length of the absorbing strip (Ω/cm).

W – The mean width of the absorbing strip and

α – The absorbing coefficient of the absorbing strip.

2. Abbot silver disk pyrliometer

This instrument was designed by *Abbot* in 1902 for the radiation measurement. A schematic diagram of this device is shown in Figure 2.8. In this pyrliometer, a silver disk painted black on its radiation-receiving side which is 3.8 cm in diameter and 0.7 cm thick is positioned at the lower end of a tube with diaphragms to limit the whole aperture to 5.7°.

The disk is suspended by three fine steel wires inside a copper box which is enclosed in a wooden box to protect the instrument from the temperature change of the surrounding. A mercury-in-glass thermometer is used to measure the temperature at the disk. A good contact between the silver disk and thermometer bulb is maintained by using mercury. The thermometer stem is bent at right angle to make this instrument more compact and easy to use. It is also supported in a metallic protective tube. A cylinder with diaphragms inside is fitted in the wooden container to let the direct solar radiation falling onto the silver disk. There is a metallic-plate shutter at the top end of the cylinder to block or allow the passage of solar radiation to the disk.

During the measurement phase, the disk is heated by solar radiation and its temperature rises. The intensity of this radiation is ascertained by measuring the temperature change of the disk between measurement phase and shading phase with the mercury-in-glass thermometer. This type is widely used for calibrating pyranometers.

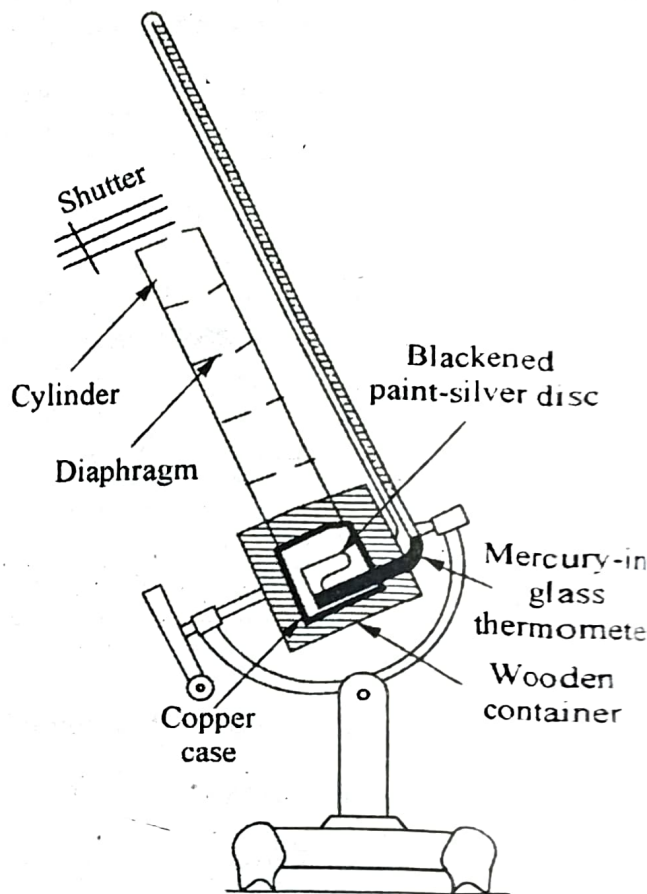


Figure 2.8 Abbot silver disk pyrhelimeter

3. Eppley pyrhelimeter

The Eppley normal incidence pyrhelimeter has found a wide acceptance in many parts of the world. It uses a thin silver disk of 9 mm in diameter as a receiver which is coated with optical black paint. Fifteen junctions of fine bismuth-silver thermocouples are in thermal contact with an electrical insulation from the lower surface of the disk. The cold junctions are in contact with the copper tube of the instrument. The unit is mounted at the base of a double-walled brass tube which is chromed externally and blackened internally. A series of limiting diaphragms limits the aperture to a circular cone of full angle of 5.7° . The brass tube is filled with dry air and sealed with a crystal quartz window which is removable. A manually rotatable disk which can accommodate three filters and leave one aperture for total spectral measurement is provided. A diopter is used to determine the direction of the sun.

This instrument is supplied with an electrically driven equatorial mount for solar tracking. The output of the pyrheliometer can be either directly recorded on a strip chart recorder or integrated over an appropriate time period and registered. This instrument is found to be very stable.

4. Linke-Feussner pyrheliometer

It is one of the most convenient instruments used for measuring the direct radiation at normal incident with and without filters. A schematic diagram of this device is shown in Figure 2.9.

The main body of the instrument consists of six massive copper rings. These rings are contoured on the inside of a tube to produce a set of radiation diaphragms for decreasing internal reflections, defining the acceptance angle of the instrument and limiting turbulent air currents inside the instrument. Thus, it secures a high stability and good sensitivity. A rotating disk with filters is positioned in the upper end of the tube.

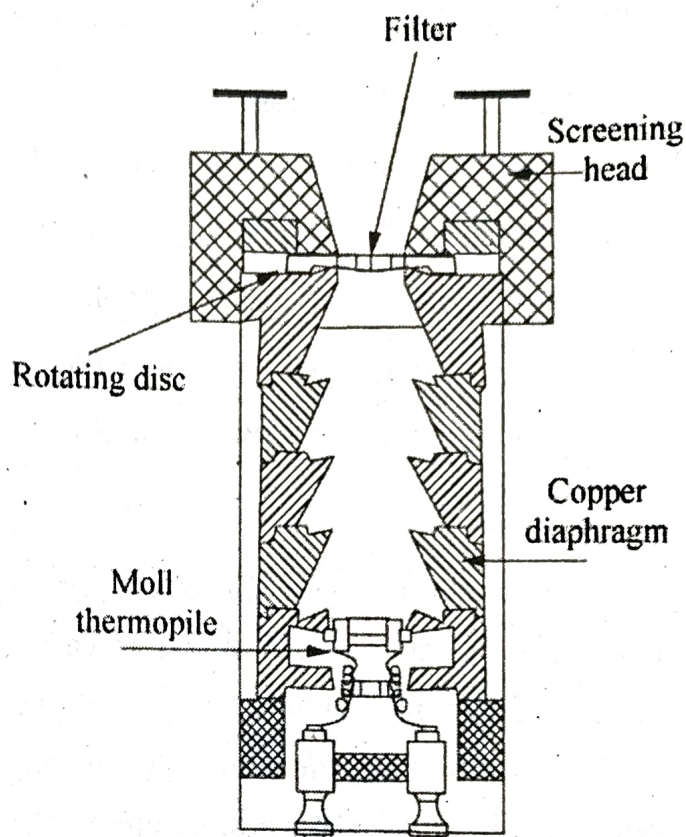


Figure 2.9 Linke-Feussner pyrheliometer

At the upper extremity of the tube is a special screening head which eliminates the unwanted reflection in filter measurements. This pyrheliometer employs a specially designed Moll thermopile receiver consisting of 40 manganese-constantan thermocouple arranged in a circle of 1 cm diameter. The thermocouples are in two equal sectional arrays. One section is exposed to the radiation being measured and the other is shaded. Thus, the sections tend to compensate each other for short period temperature fluctuations of the environment. The aperture angle is approximately 10° .

In this instrument, five filters are usually provided fitted in the rotating disk. The filter mount is as well fitted with a double walled opaque disk for the use in zeroing the instrument. This instrument is installed on a manually operated azimuth-elevation mount by which it can be oriented in any direction.